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The Wage Effects of Offshoring: Evidence from Danish Matched Worker-Firm Data

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Administrative data and sample	4
3.2	Offshoring measures	4
3.3	Export measures	4
3.4	Worker skill and task measures	5
3.5	Main outcomes	5
4	Model and empirical specifications	5
4.1	Production framework	5
4.2	Worker-level wage equation	5
4.3	Specifications with and without firm controls	6
4.4	Instrumental variables	6
5	Identification and empirical design	6
5.1	Main identification problem	6
5.2	Job-spell fixed effects	7
5.3	Why instruments are needed even with job-spell fixed effects	7
5.4	Exclusion restriction	7
5.5	First-stage relevance	7
5.6	Cohort design for total incidence	8
5.7	Main threats and responses	8
6	Tables and figures	8
6.1	Table 1: Descriptive Statistics; Table 2: Some Patterns of Offshoring and Exports	8
6.2	Table 3: Firm-Level Effects of Trade; Table 4: First-Stage FE-IV Regressions	9
6.3	Table 5: Worker-Level Wage Regressions; Table 6: Net Effect of Trade on Wages	10
6.4	Table 7: Robustness Exercises; Table 8: Wage Effects by Task Characteristics and Industry Experience	11
6.5	Table 9: Worker Mobility in Response to Rise in Offshoring; Table 10: The Overall Effects of Offshoring	12
6.6	Figure 1: The Overall Effects of Offshoring	13
7	Short summary	14

8	Conclusion	15
8.1	Core conclusion	15
8.2	Economic intuition	15
8.3	Why the identification matters	15
8.4	Implications for inequality	15
8.5	Policy implications	15
8.6	Limitations	15
8.7	Takeaway	15

1 Big picture

- **Question.** How does offshoring affect individual workers' wages, and how do these effects differ by skill, task characteristics, and worker mobility?
- **Central problem.** Offshoring is endogenous. More productive firms both offshore more and pay higher wages, so raw correlations can understate the wage losses from offshoring or even give the wrong sign.
- **Empirical setting.** The paper uses Danish administrative data that link workers to firms and match those firms to detailed imports and exports by product-country over 1995–2006.
- **Core design.** The authors instrument firm-level offshoring and exporting using external country-product shocks: world export supply, world import demand, and transport costs.
- **Main wage result.** Within job spells, offshoring lowers low-skilled workers' wages and raises high-skilled workers' wages, increasing within-firm skill premia.
- **Exports matter too.** Exporting raises wages for both low-skilled and high-skilled workers, so the net wage effect of trade depends on whether a worker's firm expands exports or offshores inputs.
- **Task heterogeneity.** Conditional on education, wage effects vary by what workers do. Routine task intensity is associated with larger offshoring wage losses, while some non-routine knowledge sets are more protected.
- **Overall effects.** Offshoring affects workers not only through wages within job spells but also through displacement and reemployment. Cohort estimates show persistent losses for low-skilled workers, especially for leavers.
- **Economic takeaway.** Offshoring is not simply “bad for all workers” or “good through productivity.” It reallocates gains and losses across workers inside firms, across skill groups, across tasks, and through employment dynamics.

2 Incremental contribution of the paper

This paper's incremental contribution is not merely that it estimates another offshoring-wage regression. Its contribution is that it connects worker-level wage outcomes, firm-level trade exposure, and product-country trade shocks in one matched framework.

1. **Matched worker-firm-trade data.** Earlier work often used industry-level trade measures. This paper observes the firm that employs each worker and that firm's imports and exports by product and country.
2. **Within-job-spell wage effects.** By using job-spell fixed effects, the paper compares the same worker-firm relationship over time. This separates wage changes from changes in workforce composition.
3. **Instrumented firm-level offshoring and exporting.** The paper constructs instruments using world supply, world demand, and transport-cost shocks at the country-product level. This directly targets the simultaneity problem that productive firms both trade more and pay more.
4. **Skill and task heterogeneity.** The paper goes beyond high-skill versus low-skill categories by showing that the wage effects of offshoring depend on occupational task characteristics.
5. **Net trade effect.** The paper estimates offshoring and exporting together, allowing the net effect of trade to differ across workers in the same skill group depending on the firm's changing trade exposure.
6. **Cohort-based total incidence.** The paper tracks worker cohorts before and after offshoring shocks to capture displacement and reemployment effects, not just wages among workers who remain with the same firm.
7. **Distributional focus.** The paper shows that trade generates within-group inequality: some low-skilled workers gain when exports expand, while some high-skilled workers lose when displacement effects dominate.

Interpretation: The paper advances the literature by moving from industry-level wage correlations to firm-worker-level causal incidence. It quantifies how the same trade shock can have different effects depending on worker skill, task content, and whether the worker stays or leaves.

3 Data and measurement

3.1 Administrative data and sample

The empirical analysis combines Danish matched employer–employee data with detailed firm-level customs data. The sample covers private-sector Danish manufacturing firms from 1995 to 2006.

- The firm-level data contain roughly 9,800 firm-year observations.
- The worker-firm data contain about 1.95 million worker-firm-year observations.
- The data track essentially the population of Danish private-sector workers, allowing workers to be followed across employers and after separation.
- The trade data record firm imports and exports by HS product code and source or destination country.

Interpretation: The richness of the data is central to the paper. Without worker-firm links, the authors could not separate within-job-spell wage effects from worker sorting. Without product-country trade flows, they could not build firm-specific trade shocks.

3.2 Offshoring measures

The paper distinguishes broad and narrow measures of offshoring.

- **Broad offshoring** includes imported intermediate inputs used by the firm.
- **Narrow offshoring** focuses on imported inputs that are closer substitutes for the firm’s own production, based on product-code proximity.
- The main wage regressions use narrow offshoring because it is closer to the task-substitution channel.

The generic measure is:

$$\log M_{jt},$$

where M_{jt} is firm j ’s offshoring level in year t .

Interpretation: Narrow offshoring is the most economically relevant measure for wages because it is more likely to represent tasks that could otherwise have been performed by the firm’s own workers.

3.3 Export measures

The paper also measures firm exports:

$$\log X_{jt},$$

where X_{jt} is firm j ’s export value in year t .

- Exporting may raise labor demand through scale effects.
- Offshoring may reduce labor demand for workers whose tasks are substitutes for imported inputs.
- Estimating both together is essential because firms often offshore and export at the same time.

Interpretation: A regression that includes only offshoring would conflate the wage-reducing substitution effect of imports with the wage-increasing scale effect of exports.

3.4 Worker skill and task measures

Workers are divided into high-skilled and low-skilled groups. The paper also uses occupational task measures from O*NET.

- High-skilled workers are those with higher educational attainment.
- Low-skilled workers are the complement.
- Task dimensions include routine intensity, non-routine content, mathematics, social sciences, natural sciences, communication, and industry-specific experience.

Interpretation: Skill categories capture broad education differences. Task measures capture whether a worker’s actual work is easier or harder to substitute with imported inputs.

3.5 Main outcomes

The paper studies both firm-level and worker-level outcomes.

- Firm outcomes include employment, output, profits, capital per worker, average wage bill, material inputs, and domestic inputs.
- Worker outcomes include log hourly wage, log annual labor income, job spell status, mobility, and cohort-level wages and earnings.
- The paper distinguishes wages within job spells from overall earnings effects after displacement.

Interpretation: The paper’s wage incidence analysis has two layers. The first asks what happens to wages when a worker stays in a job spell. The second asks what happens to the worker’s total income path when offshoring may also trigger separation.

4 Model and empirical specifications

4.1 Production framework

The theoretical motivation begins with a production function in which output depends on productivity, capital, skilled labor, and a composite of unskilled labor and imported inputs:

$$Y_{jt} = A_{jt} K_{jt}^{\beta} H_{jt}^{\alpha} C_{jt}^{1-\alpha-\beta}.$$

The composite input C_{jt} combines unskilled labor and imported inputs.

Interpretation: If imported inputs substitute for unskilled labor, offshoring reduces demand for low-skilled labor. If imported inputs raise productivity or output scale, they can raise labor demand. The sign is therefore theoretically ambiguous.

4.2 Worker-level wage equation

The wage equation is motivated by firm labor demand and upward-sloping firm-specific labor supply:

$$\log w_{ijt} = b_{LM} \log M_{jt} + b_{HM} S_i \log M_{jt} + b_{LX} \log X_{jt} + b_{HX} S_i \log X_{jt} + X'_{it} \beta + Z'_{jt} \gamma + \alpha_{ij} + \phi_{rt} + \phi_{It} + \varepsilon_{ijt}.$$

Here:

- w_{ijt} is worker i 's wage at firm j in year t ;
- M_{jt} is offshoring;
- X_{jt} is exports;
- S_i is a high-skill indicator;
- α_{ij} is a job-spell fixed effect;
- ϕ_{rt} and ϕ_{It} are region-year and industry-year fixed effects.

Interpretation: The offshoring coefficient gives the low-skilled wage elasticity. The sum of the low-skilled coefficient and the high-skill interaction gives the high-skilled wage elasticity.

4.3 Specifications with and without firm controls

The paper estimates versions with firm-level controls and versions without them.

- With firm controls, the coefficient is closer to the direct wage effect holding firm scale and factor mix fixed.
- Without firm controls, the coefficient also allows productivity and scale channels to operate.

Interpretation: Comparing these specifications helps separate direct substitution effects from broader firm-level productivity and output effects.

4.4 Instrumental variables

The paper instruments offshoring and exports using external country-product shocks.

For imports/offshoring:

- world export supply;
- transport costs for imports.

For exports:

- world import demand;
- transport costs for exports.

The first-stage logic is:

$$\begin{aligned}\log M_{jt} &= \Pi_M Z_{jt}^M + controls + u_{jt}, \\ \log X_{jt} &= \Pi_X Z_{jt}^X + controls + v_{jt}.\end{aligned}$$

Interpretation: The instruments shift the cost or attractiveness of trading with particular countries and products but are designed to be external to Danish firms' own productivity shocks.

5 Identification and empirical design

5.1 Main identification problem

Firms that offshore are systematically different. They are larger, more productive, more capital-intensive, and pay higher wages. A simple wage regression may therefore confuse firm productivity shocks with offshoring effects.

Interpretation: The central identification problem is simultaneity: the same unobserved firm shock can raise wages, offshoring, and exports at the same time.

5.2 Job-spell fixed effects

The paper uses job-spell fixed effects, comparing wage changes for the same worker-firm match over time.

α_{ij} absorbs permanent worker-firm match quality.

- This controls for stable worker ability.
- It controls for stable firm wage premia.
- It controls for match-specific wage components.

Interpretation: This is stronger than separate worker and firm fixed effects because it does not rely on worker mobility to identify firm wage premia. It asks how wages change within continuing employment relationships.

5.3 Why instruments are needed even with job-spell fixed effects

Job-spell fixed effects remove fixed heterogeneity but not time-varying productivity or demand shocks.

- A productivity shock can increase imported input use.
- The same shock can increase wages.
- A demand shock can increase exports and wages.

Interpretation: The IV design isolates changes in offshoring and exports driven by foreign supply, foreign demand, and transport costs rather than by the Danish firm’s own productivity.

5.4 Exclusion restriction

The exclusion restriction is that world export supply, world import demand, and transport-cost shocks affect Danish worker wages only through their effects on the firm’s offshoring and exports.

$$Z_{jkt} \rightarrow \log M_{jt}, \log X_{jt} \rightarrow \log w_{ijt}.$$

Interpretation: The paper’s case is strongest for offshoring instruments because world export supply and transport costs are plausibly external to Danish firms. The export instruments may be more vulnerable to correlated demand shocks, and the authors explicitly note that the export estimates require more caution.

5.5 First-stage relevance

Table 4 shows that the excluded instruments predict offshoring, exports, and their skill interactions. The F-statistics are generally large enough to support relevance.

Interpretation: Strong first stages matter because the worker-level IV estimates rely on predicted trade changes. Weak instruments would make the wage estimates unstable and hard to interpret.

5.6 Cohort design for total incidence

The within-job-spell regressions omit workers who leave the firm. To capture total incidence, the paper forms cohorts of workers employed by the firm before an offshoring shock and tracks them over time.

$$\Delta \log y_{iv+k} = \alpha_v + \alpha_i + \sum_{k=1}^5 \beta_k \Delta \log OFF_{jv+1} + controls + \varepsilon_{iv+k}.$$

- The cohort is fixed before the offshoring shock.
- Workers are followed whether they stay, leave, become unemployed, or reattach elsewhere.
- The design estimates both wage and displacement effects.

Interpretation: This is essential because offshoring can harm workers through job loss even if within-spell wages for some stayers rise.

5.7 Main threats and responses

- **Endogenous input choice:** firms may switch inputs after productivity shocks. The instruments use external country-product shocks and pre-existing product-country patterns.
- **Composition bias:** firms may retain only better workers. The job-spell design and cohort design address different parts of this problem.
- **Export demand confounds:** world import demand could correlate with demand for Danish output. Export estimates are interpreted more cautiously.
- **Selection through mobility:** leavers differ from stayers. Table 9 and the cohort analysis directly examine this channel.
- **Skill and task sorting:** workers in different tasks may be on different trends. Task analyses condition on skill and job-spell structure, but remain more descriptive than the main IV wage estimates.

Interpretation: The paper does not rely on one specification. It combines firm outcomes, first-stage evidence, worker-level IV wage regressions, robustness tests, task heterogeneity, and cohort incidence estimates.

6 Tables and figures

6.1 Table 1: Descriptive Statistics; Table 2: Some Patterns of Offshoring and Exports

Read:

- Table 1 shows that the firm sample contains about 9,800 firm-year observations and the worker sample contains about 1.95 million worker-firm-year observations.
- Average narrow offshoring is about 12 percent of gross output and 27 percent of material purchases.
- Exports are large: average exports are about 45 percent of gross output.
- Table 2 shows that imports and exports are concentrated in a few product categories; the top five import products account for 92.1 percent of imports.
- The pre-sample flows explain a large share of in-sample trade flows, supporting the use of pre-existing input and export patterns in the instruments.

Detailed interpretation: These tables establish two facts required for the research design. First, Danish manufacturing firms are deeply engaged in trade, so trade shocks are economically meaningful. Second, firms trade in narrow and stable product-country categories, so external shocks to those markets can generate firm-specific variation.

Economic message: The data are rich enough to link specific trade shocks to specific firms and then to specific workers. This is the empirical foundation for the whole paper.

TABLE 1—DESCRIPTIVE STATISTICS

	Observations	Mean	SD
<i>Firm-level data</i>			
log employment	9,820	4.94	0.89
log gross output	9,804	18.89	1.05
log capital per worker	9,759	12.39	0.98
log average wage bill per worker	9,772	12.54	0.22
log accounting profits	7,816	9.07	1.70
High-skill share	9,772	0.16	0.12
Low-skill share	9,772	0.84	0.12
<i>Firm-level trade data</i>			
log(broad offshoring)	9,820	16.85	1.53
Broad offshoring/gross output	9,804	0.19	0.16
Broad offshoring/material purchases	9,756	0.43	0.29
Broad offshoring, log deviation from firm mean	9,820	0.49	0.57
log(narrow offshoring)	9,249	16.00	2.26
Narrow offshoring/gross output	9,804	0.12	0.15
Narrow offshoring/material purchases	9,756	0.27	0.28
Narrow offshoring, log deviation from firm mean	9,249	0.82	0.94
log(exports)	9,555	17.54	2.06
Exports/gross output	9,804	0.45	0.32
Exports, log deviation from firm mean	9,555	0.46	0.66
<i>Worker-firm data</i>			
Hourly wage	1,950,896	192.85	70.19
log hourly wage	1,950,896	5.19	0.31
log gross output	1,950,896	20.50	1.69
log employment	1,950,896	6.44	1.49
log capital per worker	1,950,896	12.59	0.89
High-skill	1,950,896	0.19	0.14
Experience	1,950,896	17.93	9.31
Union	1,950,896	0.88	0.33
Married	1,950,896	0.59	0.49

Notes: The data used for the last panel titled “Worker-firm data” has worker-firm-year observations, and the data used for the other panels has firm-year observations. For each variable we report its mean and standard deviation across all observations.

TABLE 2—SOME PATTERNS OF OFFSHORING AND EXPORTS (*in percent*)

<i>Share of import value</i>	
Raw materials	7.8
Machinery and machinery parts	16.9
Narrow offshoring, same HS2 as sales	87.4
Narrow offshoring, same HS4 as sales	70.8
<i>Share of trade</i>	
Top 2 products in imports	67.9
Top 5 products in imports	92.1
Top 2 products in exports	51.3
Top 5 products in exports	77.0
<i>Pre-sample flows</i>	
In-sample share of imports	64.4
In-sample share of exports	77.7

Notes: The data used for Table 2 has firm-year-product-source-country observations for import flows, and firm-year-product-destination-country observations for export flows. The panel titled “Share of import value” shows the fractions of total import value accounted for by various categories; e.g. the sum of raw-materials imports across all observations is 7.8 percent of the sum of import values across all observations in our sample. The “Share of trade” panel shows the fractions of total import/export value accounted for by the top 2/5 products of a given firm-year. The “Pre-sample flows” panel shows the fractions of total import/export values accounted for by the firm-product-country flows that appear in the pre-sample years.

(a) **Table 1: Descriptive Statistics.** Original table/figure cropped from the paper.

(b) **Table 2: Some Patterns of Offshoring and Exports.** Original table/figure cropped from the paper.

6.2 Table 3: Firm-Level Effects of Trade; Table 4: First-Stage FE-IV Regressions

Read:

- Table 3 shows that offshoring firms are larger, more productive, more profitable, and pay higher average wages in raw comparisons.
- With firm fixed effects, rising offshoring is still positively correlated with output and wages, but predicted offshoring is negatively related to employment when exports are included.
- Predicted exports are strongly positively associated with employment, output, profits, and domestic material inputs.
- Table 4 shows the first-stage relationships between instruments and the endogenous trade variables.
- World export supply and transport costs predict offshoring, while world import demand and transport costs predict exports.

Detailed interpretation: The comparison between Tables 3 and 4 clarifies why identification is difficult. Raw offshoring is associated with high wages because good firms offshore. The first-stage IV strategy strips out this endogenous component by using external trade-cost and world-market shocks.

How it fits the paper: Table 3 motivates the endogeneity problem. Table 4 verifies that the excluded instruments have enough predictive power to support the wage regressions.

TABLE 3—FIRM-LEVEL EFFECTS OF TRADE

	OLS	Firm FE	Firm FE, predicted offshoring	Firm FE predicted offshoring and exports
	Offshorer dummy (1)	log(offshoring) (2)	log(offshoring) (3)	log(offshoring) (4)
log employment	0.681***	0.044***	−0.106**	−0.205***
log gross output	0.958***	0.082***	0.394***	0.511***
log accounting profits	0.953***	0.066***	0.506***	0.001
log(capital per worker)	0.161*	0.005	0.245***	0.119***
log(wage bill per worker)	0.040**	0.014***	0.224***	0.131***
log material inputs	1.162***	0.083***	0.195**	−0.140*
log domestic material inputs	0.668	0.037***	0.355***	−0.082
Share of high-skilled workers	−0.007	0.002*	0.091***	0.053***
Materials/output	0.093***	0.005**	−0.049*	−0.063**
Domestic materials/output	−0.043**	−0.011***	0.012	−0.026

Notes: The cells are coefficient estimates of various regressions, whose dependent variables are down the rows and regressors are along the columns. For example, column 1 shows that when we regress log(employment) on the offshorer dummy ($=1$ if firm j has positive offshoring value in year t), we get a coefficient of 0.681 (significant). The data used for column 1 includes all firm-years, that for two and three only the firm-years that have positive offshoring values, and that for four and five only the firm-years with positive offshoring and export values (which corresponds to our main estimation sample). In columns 3–5 we correct the standard errors for the fact that the covariates are estimated, following Wooldridge (2002).

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

(a) **Table 3: Firm-Level Effects of Trade.** Original table/figure cropped from the paper.

TABLE 4—FIRST-STAGE FE-IV REGRESSIONS

	log(offshoring)		× high skill		log(exports)		× high skill	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log WES, offshoring	0.1928*** [2.70]	0.3087*** [4.38]	−0.0428*** [−5.74]	−0.0255*** [−3.61]	0.0068 [0.08]	0.0869 [1.02]	−0.0232*** [−3.28]	−0.0106 [−1.44]
log transport costs, offshoring	−17.1988*** [−3.04]	−19.7103*** [−3.41]	0.2078 [0.40]	−0.3583 [−0.75]	3.4490 [1.35]	1.0740 [0.40]	0.7999** [2.39]	0.3821 [1.34]
log WID, exports	−0.0839 [−0.66]	0.0778 [0.62]	−0.0541*** [−5.37]	−0.0253*** [−3.03]	0.2346*** [3.12]	0.3606*** [4.67]	−0.0339*** [−4.23]	−0.0125** [−2.03]
log transport costs, exports	22.1646*** [3.18]	23.5056*** [3.34]	−2.7704*** [−2.98]	−2.1234** [−2.50]	−7.8695 [−1.33]	−5.8672 [−0.97]	−0.6316 [−0.96]	−0.0246 [−0.04]
<i>Interactions with high skill dummy</i>								
log WES, offshoring	−0.0564 [−0.83]	−0.0827 [−1.29]	0.3551*** [4.66]	0.3495*** [4.62]	0.1086* [1.75]	0.0841 [1.33]	0.2684*** [5.30]	0.2646*** [5.13]
log transport costs, offshoring	1.0166 [0.27]	−0.6953 [−0.18]	−17.6820*** [−3.09]	−18.288*** [−3.19]	2.6859 [1.12]	0.8210 [0.32]	1.3016 [0.37]	0.8182 [0.23]
log WID, exports	0.0519 [0.58]	0.1489 [1.61]	0.3585*** [4.55]	0.3750*** [4.74]	−0.1898*** [−3.90]	−0.1143** [−2.32]	0.3374*** [6.35]	0.3494*** [6.52]
log transport costs, exports	−2.7041 [−0.77]	−4.4856 [−1.25]	26.2738*** [4.42]	25.9308*** [4.35]	−3.0533 [−1.03]	−4.3535 [−1.44]	−5.6607* [−1.80]	−5.9273* [−1.88]
Additional firm controls	Yes	No	Yes	No	Yes	No	Yes	No
Additional worker controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,928,599	1,928,599	1,928,599	1,928,599	1,950,896	1,950,896	1,950,896	1,950,896
Number of job-spell fixed effects	383,035	383,035	383,035	383,035	384,257	384,257	384,257	384,257
R^2	0.146	0.107	0.074	0.063	0.207	0.157	0.074	0.059
F -statistics for instruments	4.379	30.040	5.300	19.330	8.618	22.360	10.140	15.740

Notes: Table 4 presents the first stage regressions for log offshoring, log exports, and their skill interactions, using world export supply (WES), world import demand (WID), and transport costs as excluded instruments. Only these excluded instruments are reported. All specifications include job-spell, industry-year, and regional fixed effects. Additional firm controls are log output, log employment, log capital-labor ratio, and the share of high-skilled workers. Additional worker controls include experience, experience squared, and dummies for union membership and marriage. Robust t -statistics in brackets. Standard errors clustered at firm-year levels.

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

(b) **Table 4: First-Stage FE-IV Regressions.** Original table/figure cropped from the paper.

6.3 Table 5: Worker-Level Wage Regressions; Table 6: Net Effect of Trade on Wages

Read:

- Table 5 shows that IV estimates are much larger in magnitude than fixed-effect estimates.
- In the preferred FE-IV specification, offshoring lowers low-skilled hourly wages by roughly 2.2 percent for a 100 percent increase in offshoring.
- The interaction with high skill is positive and large, so the total effect for high-skilled workers is positive.
- Exports raise wages for both skill groups, with wage elasticities around five percent in the main hourly wage specifications.
- Table 6 combines the offshoring and export coefficients to show that net wage effects vary substantially even within the same skill group.

Detailed interpretation: Table 5 contains the key causal wage estimates. Table 6 shows why a single “trade effect” is misleading: some firms increase offshoring and exports simultaneously, and the wage effect depends on the mix of those changes.

Economic message: Offshoring redistributes income between skill groups inside firms, while exporting raises demand more broadly. The net effect of globalization is therefore worker- and firm-specific.

	log hourly wage				log labor income	
	FE		FE-IV		FE-IV	
	(1)	(2)	(3)	(4)	(5)	(6)
log offshoring	−0.0025** [−2.43]	−0.0014 [−1.41]	−0.0222** [−2.56]	−0.0228*** [−3.70]	−0.0148** [−2.06]	−0.0100* [−1.94]
log offshoring × high skilled	0.0060*** [5.59]	0.0061*** [5.57]	0.0061*** [7.71]	0.0523*** [7.78]	0.0220*** [3.59]	0.0239*** [3.84]
log exports	0.0044** [2.02]	0.0060*** [2.82]	0.0493*** [4.48]	0.0531*** [7.63]	0.0391*** [3.44]	0.0538*** [6.47]
log exports × high skilled	−0.0005 [−0.25]	0.0000 [0.02]	0.0008 [0.08]	0.0019 [0.18]	0.0364*** [3.62]	0.0398*** [3.83]
log output	0.0141*** [4.56]		0.0064 [0.88]		0.0071 [0.87]	
log employment	0.0158*** [4.62]		−0.0043 [−0.39]		0.0251** [2.51]	
log capital-labor ratio	0.0039*** [3.16]		0.0051*** [4.15]		0.0009 [0.73]	
Share, high-skilled workers	0.0879*** [5.07]		0.1436*** [6.66]		0.1361*** [5.14]	
Experience	0.0171*** [13.27]	0.0181*** [14.00]	0.0156*** [11.58]	0.0157*** [11.42]	0.2186*** [34.43]	0.2188*** [34.33]
Experience ² × 100	−0.05*** [−84.53]	−0.05*** [−84.77]	−0.05*** [−80.68]	−0.05*** [−77.89]	−0.11*** [−95.33]	−0.11*** [−94.18]
Union	0.0142*** [13.38]	0.0141*** [13.29]	0.0156*** [14.29]	0.0153*** [14.49]	0.0301*** [21.34]	0.0301*** [21.59]
Married × 100	0.34*** [6.42]	0.35*** [6.59]	0.29*** [5.43]	0.29*** [5.39]	−0.80*** [−8.92]	−0.81*** [−8.96]
Observations	1,928,599	1,928,599	1,928,599	1,928,599	1,928,428	1,928,428
Number of job-spell fixed effects	383,035	383,035	383,035	383,035	383,033	383,033
R ²	0.155	0.153	0.155	0.153	0.103	0.101

Notes: Table 5 presents the results from worker-level Mincer regressions, using either log hourly wage or log annual labor income as dependent variables. All specifications include job-spell, industry-year, and regional fixed effects. log offshoring, log exports, and their skill interactions are instrumented using world export supply (WES), world import demand (WID) and transport cost in the FE-IV columns. Robust *t*-statistics in brackets. Standard errors clustered at firm-year levels. We report 100 times the coefficient estimates for experience² and marital status.
***Significant at the 1 percent level.

(a) Table 5: Worker-Level Wage Regressions. Original table/figure cropped from the paper.

TABLE 6—NET EFFECT OF TRADE ON WAGES

		Annual percent-change in exports				
		Min	−30	0	30	
		−30	0	30	Max	
<i>Panel A. Low-skilled workers</i>						
Annual	Min	−30	3.1	6.7	5.6	2.2
percent-change			−0.92	0.73	2.06	6.60
in offshoring	−30	0	1.9	13.7	9.7	1.4
			−2.12	−0.17	0.74	2.94
	0	30	1.6	9.8	15.2	2.8
			−2.90	−0.78	0.25	2.14
	30	Max	1.5	5.3	13.3	6.3
			−6.15	−1.96	−0.72	1.44
<i>Panel B. High-skilled workers</i>						
Annual	Min	−30	3.2	6.3	4.5	1.4
percent-change			−7.08	−2.38	−1.00	0.56
in offshoring	−30	0	1.6	15.1	10.9	1.4
			−3.10	−0.66	0.17	1.93
	0	30	1.8	11.3	16.4	2.8
			−2.16	−0.05	0.95	2.91
	30	Max	1.3	5.0	12.4	4.6
			−1.28	1.11	2.20	5.68

Notes: Table 6 presents the net effect of trade, including both offshoring and exports, on worker-level hourly wages. For panel A we group the firm-year observations into 16 bins according to the annual changes in offshoring and export values, and these bins correspond to the 16 cells. For each cell we report two numbers. The bold-faced number is the predicted median hourly wage change for low-skilled workers (calculated using the wage-elasticity estimates from Table 5), and the normal-fonted figure is the fraction of low-skilled workers in this cell. For example, for the firm-year observations for which offshoring decreases by more than 30 percent (first row) and exports increase by more than 30 percent (last column), the median predicted hourly wage for low-skilled workers rises by 6.60 percent, and these low-skilled workers account for 2.2 percent of the low-skilled workforce in our sample. Panel B is similarly structured for high-skilled workers.

(b) Table 6: Net Effect of Trade on Wages. Original table/figure cropped from the paper.

6.4 Table 7: Robustness Exercises; Table 8: Wage Effects by Task Characteristics and Industry Experience

Read:

- Table 7 shows that the main wage patterns survive restricting to long job spells, high-income countries, moving averages of instruments, and dropping computers and building supplies.
- The offshoring coefficient remains negative for low-skilled workers and the high-skill interaction remains positive.
- Table 8 adds task characteristics and industry-specific experience.
- Routine task intensity is associated with larger wage losses from offshoring.
- Some non-routine or knowledge-intensive dimensions interact positively with offshoring, indicating that task content matters even within education groups.

Detailed interpretation: Table 7 strengthens the credibility of the main estimates by showing that they are not driven by one sample choice or one instrument construction. Table 8 gives the mechanism more texture: the wage impact depends not only on education but also on whether a worker’s tasks are substitutable by imported inputs.

How it fits the paper: These tables move from “does offshoring affect wages?” to “which workers are most exposed to offshoring?”

TABLE 7—ROBUSTNESS EXERCISES

Robustness check:	log hourly wage							
	I. 7+ year job spells		II. High-income countries		III. 3-year MAs (WID/WES)		IV. Drop computers and building supplies	
	FE-IV	FE-IV	FE-IV	FE-IV	FE-IV	FE-IV	FE-IV	FE-IV
log(offshoring)	-0.0105 [-1.64]	-0.0155*** [-2.83]	-0.0168** [-2.53]	-0.0190*** [-3.67]	-0.0177** [-2.14]	-0.0181*** [-3.01]	-0.0354*** [-3.44]	-0.0317*** [-4.53]
log(offshoring) × high-skilled	0.0510*** [7.06]	0.0536*** [7.24]	0.0271*** [4.94]	0.0269*** [4.88]	0.0519*** [7.49]	0.0530*** [7.47]	0.0833*** [9.57]	0.0892*** [9.92]
log(exports)	0.0665*** [6.21]	0.0621*** [7.46]	0.0464*** [4.52]	0.0492*** [7.14]	0.0478*** [4.77]	0.0503*** [6.87]	0.0488*** [2.90]	0.0656*** [7.18]
log(exports) × high-skilled	0.0055 [0.48]	0.0026 [0.22]	0.0344*** [3.87]	0.0370*** [4.28]	0.0057 [0.54]	0.0060 [0.57]	-0.0436*** [-3.20]	-0.0461*** [-3.25]
First stage IV F-statistics:								
log offshoring	7.21	13.42	6.25	11.62	4.46	9.13	4.76	8.33
... × high skill	28.49	24.10	22.41	17.53	22.56	21.05	26.21	18.97
log exports	5.79	14.92	5.35	10.43	4.98	10.30	3.80	7.92
... × high skill	17.54	16.11	18.45	15.36	16.60	16.52	18.90	14.70
Other firm-level controls	Yes	No	Yes	No	Yes	No	Yes	No
Observations	967,053	967,053	1,917,625	1,917,625	1,928,599	1,928,599	1,692,736	1,692,736
No. job spells	103,989	103,989	380,781	380,781	383,035	383,035	338,922	338,922
R ²	0.186	0.183	0.155	0.153	0.155	0.153	0.156	0.155

Notes: Table 7 presents the results from worker-level Mincer regressions, using log hourly wage as the dependent variable. All specifications include job spell, industry-year, and regional fixed effects. log offshoring, log exports, and their skill interactions are instrumented using world export supply (WES), world import demand (WID), and transport costs. "MAS" stand for moving-averages. Robust t-statistics in brackets. Standard errors clustered at firm-year levels.
 ***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

TABLE 8—WAGE EFFECTS BY TASK CHARACTERISTICS AND INDUSTRY EXPERIENCE

Interaction term (Z):	Routine	Non-routine	Non-routine (other than math)	Social sciences	Natural sciences	Communication	Industry-specific experience
log(offshoring)	0.0075 [0.83]	-0.0034 [-0.40]	-0.0055 [-0.66]	-0.0049 [-0.54]	-0.0155* [-1.70]	0.0030 [0.34]	-0.0248*** [-2.85]
log(offshoring) × high-skilled	-0.0081 [-1.20]	-0.0264*** [-3.89]	-0.0227*** [-3.37]	0.0151** [2.40]	0.0465*** [6.88]	-0.0081 [-1.28]	0.0512*** [8.53]
log(offshoring) × Z	-0.0422*** [-17.36]	0.0497*** [14.10]	-0.0443*** [-13.56]	0.0377*** [16.09]	-0.0008 [-0.44]	0.0446*** [17.23]	-0.0002 [-1.12]
log(exports)	0.0588*** [5.25]	0.0238** [2.04]	0.0147 [1.25]	0.0557*** [4.96]	0.0467*** [4.21]	0.0424*** [3.78]	0.0374*** [3.37]
log(exports) × high-skilled	-0.0002 [-0.02]	0.0140 [1.41]	0.0201** [2.04]	0.0003 [0.03]	0.0079 [0.78]	0.0071 [0.73]	-0.0028 [-0.30]
Other firm-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,570,088	1,570,088	1,570,088	1,570,088	1,570,088	1,570,088	1,906,704
Number of job spells	376,590	376,590	376,590	376,590	376,590	376,590	378,474
R ²	0.142	0.143	0.143	0.142	0.141	0.143	0.155

Notes: t-statistics in brackets. Standard errors clustered at firm-year levels. Industry-time, regional, and job-spell fixed effects included in all specifications. Industry experience measured as the number of years in the 2-digit NACE industry between 1988 and the beginning of the job spell, and this value remains unchanged during the job spell. Task characteristics measured as the number of standard deviations from the mean. Coefficient estimates of the other variables not reported to save space.
 ***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

Instead of only examining workers by educational attainment we also examine them by

(a) **Table 7: Robustness Exercises.** Original table/figure cropped from the paper.

(b) **Table 8: Wage Effects by Task Characteristics and Industry Experience.** Original table/figure cropped from the paper.

6.5 Table 9: Worker Mobility in Response to Rise in Offshoring; Table 10: The Overall Effects of Offshoring

Read:

- Table 9 shows that workers who leave after positive offshoring shocks are younger, less experienced, and lower paid than the overall worker pool.
- This indicates selection into separation, especially among low-skilled workers.
- Table 10 tracks worker cohorts and estimates the overall effects of offshoring on wages and earnings.
- Low-skilled workers experience persistent wage and earnings losses; the five-year PDV for low-skilled wages is about -11.55 percent.
- First-year leavers suffer very large earnings losses, with five-year PDVs around -50 percent for both skill groups.

Detailed interpretation: Table 9 is a warning: within-job-spell wage regressions miss a selected group of leavers. Table 10 corrects the incidence perspective by following all workers from the pre-shock cohort, whether they stay or leave.

Economic message: The full cost of offshoring is not only a wage change for stayers. Displacement is a major channel, and it is especially severe for first-year leavers.

TABLE 9—WORKER MOBILITY IN RESPONSE TO RISE IN OFFSHORING

Variables:	High-skilled		Low-skilled	
	All	Leavers	All	Leavers
Hourly wage rate	244.03	235.19	173.49	167.27
Experience	15.17	12.51	18.00	15.34
Tenure	4.78	3.17	6.23	4.53
Union status	0.70	0.70	0.91	0.90
Age	39.21	36.69	40.20	37.65
Female	0.32	0.32	0.34	0.35
Married	0.63	0.56	0.58	0.51
Observations	75,964	9,045	358,399	39,429

Notes: Table 9 compares the pre-shock characteristics of the workers who separate from their employers (“Leavers”) after a positive offshoring shock with those of the full set of workers (“All”), including the leavers and those who stay. To be specific, an offshoring shock is an increase in predicted offshoring of more than 10 percent annually, where the predicted values are obtained as in Section IV. For each firm with a positive offshoring shock we consider the workers employed by this same firm two years prior to the shock.

(a) Table 9: Worker Mobility in Response to Rise in Offshoring. Original table/figure cropped from the paper.

TABLE 10—THE OVERALL EFFECTS OF OFFSHORING

	All workers							
	Wages		Earnings		Gross earnings		Earnings	
	High-skilled	Low-skilled	High-skilled	Low-skilled	High-skilled	Low-skilled	High-skilled	Low-skilled
Offs. shock	-0.0097*	-0.0274***	0.0144***	0.0011	0.0118**	-0.0029	-0.1492***	-0.1621***
year 1	[-1.76]	[-3.61]	[2.89]	[0.29]	[2.51]	[-0.80]	[-6.29]	[-6.16]
Offs. shock	-0.0126**	-0.0394***	0.0085	-0.0152***	0.0082	-0.0133***	-0.1744***	-0.1677***
year 2	[-2.22]	[-5.95]	[1.56]	[-3.72]	[1.59]	[-3.48]	[-6.96]	[-6.85]
Offs. shock	-0.0047	-0.0253***	0.0022	-0.0188***	0.0026	-0.0152***	-0.1109***	-0.0930***
year 3	[-0.74]	[-4.85]	[0.41]	[-5.46]	[0.52]	[-5.20]	[-4.68]	[-4.24]
Offs. shock	-0.0012	-0.0197***	0.0082	-0.0124***	0.0082	-0.0109***	-0.0622**	-0.0816***
year 4	[-0.19]	[-3.49]	[1.42]	[-3.61]	[1.51]	[-3.56]	[-2.56]	[-3.91]
Offs. shock	0.0150**	-0.0109*	0.0203***	0.0005	0.0191***	-0.0011	-0.0468*	-0.0689***
year 5	[2.07]	[-1.86]	[3.06]	[0.11]	[3.04]	[-0.32]	[-1.76]	[-3.09]
Five year PDV	-0.0144	-0.1155***	0.0492**	-0.0415**	0.0457**	-0.0404***	-0.5147***	-0.5408***
PDV	[0.64]	[4.89]	[2.26]	[2.96]	[2.22]	[3.13]	[5.34]	[5.81]
Observations	92,0291	4,280,135	920,303	4,280,135	920,303	4,280,135	96,366	419,415

Notes: This table reports the overall effects of offshoring, including both wages and displacement, on workers’ present and future income streams using regression (6). All regressions include cohort fixed effects, year fixed effects, and the change in log predicted exports and the change in average experience as control variables. First-year leavers are workers who are no longer employed by the firm in the year after the base year. The five-year PDV is the discounted sum of coefficients using a 4 percent discount rate. The regressions are weighted by the base year cohort size and these weights are reflected in the reported observation numbers. *t*-statistics in square brackets. Standard errors clustered at firm-year levels. The reported standard errors for the five year PDVs are calculated using the delta method.

***Significant at the 1 percent level.

(b) Table 10: The Overall Effects of Offshoring. Original table/figure cropped from the paper.

6.6 Figure 1: The Overall Effects of Offshoring

Read:

- The top panel plots wage effects for all workers over five years after an offshoring shock.
- Low-skilled workers experience persistent negative wage effects.
- High-skilled workers show small and eventually positive wage effects.
- The bottom panel plots earnings effects for first-year leavers.
- Leavers experience large earnings losses immediately, with gradual but incomplete recovery.

Detailed interpretation: Figure 1 visually summarizes the distinction between within-job wage effects and total worker incidence. The wage effects for all workers are moderate, but the earnings effects for displaced workers are large.

Economic message: Offshoring creates both winners and losers. The same shock can raise wages for high-skilled stayers while imposing large losses on displaced workers.

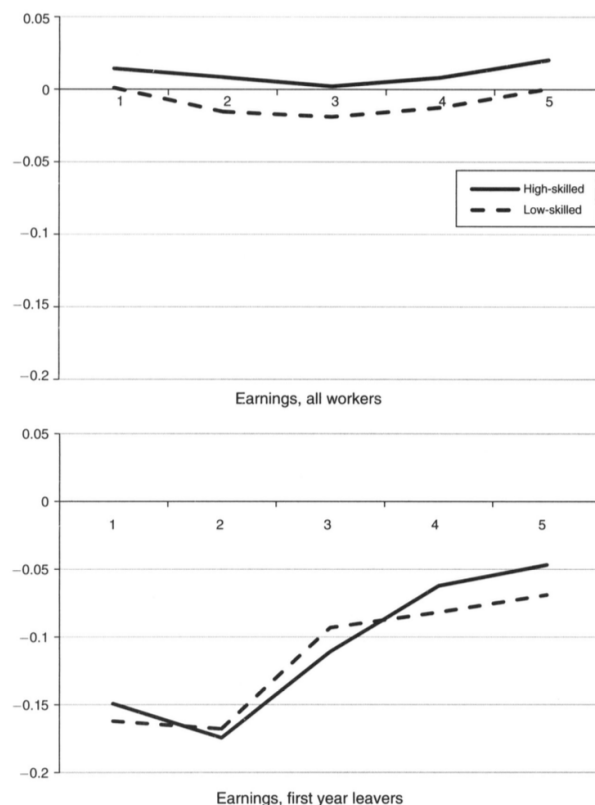


FIGURE 1. THE OVERALL EFFECTS OF OFFSHORING

Figure 1: The Overall Effects of Offshoring. Original figure/table cropped from the paper.

7 Short summary

- The paper studies how firm-level offshoring and exporting affect Danish workers' wages and earnings.
- The data link virtually all Danish workers to employers and match firms to product-country import and export flows.
- The main empirical challenge is that productive firms both offshore more and pay higher wages.
- The paper instruments offshoring and exporting with external world supply, world demand, and transport-cost shocks.
- Within job spells, offshoring lowers low-skilled wages and raises high-skilled wages.
- Exporting raises wages for both skill groups.
- The net wage effect of trade varies widely across workers because firms differ in their offshoring and export changes.
- Routine workers face larger wage losses from offshoring, while some non-routine and knowledge-intensive workers are more protected.
- Worker mobility matters: leavers after offshoring shocks have lower pre-shock wages and experience large earnings losses.
- Cohort estimates show persistent low-skilled losses, especially through displacement.
- The paper's main lesson is that trade affects workers through both within-firm wage bargaining and employment transitions.

8 Conclusion

8.1 Core conclusion

Hummels, Jørgensen, Munch, and Xiang show that offshoring has sharply heterogeneous wage effects. It lowers the wages of low-skilled workers within job spells, raises the wages of high-skilled workers, and imposes large losses on displaced workers. Exporting offsets some of these losses by raising labor demand, but the net effect of trade differs substantially across workers.

8.2 Economic intuition

The economic intuition is that imported inputs can substitute for some domestic tasks while complementing other workers and raising firm productivity. Low-skilled or routine workers are more likely to perform tasks that imported inputs can replace. High-skilled workers may benefit because offshoring expands production, increases productivity, or raises demand for complementary skills. Exporting raises scale and tends to increase wages more broadly.

8.3 Why the identification matters

The paper’s IV strategy is crucial because the raw data make offshoring look less harmful. Productive firms both offshore and pay more, so ordinary fixed-effect estimates are biased toward zero or positive values. External trade shocks reveal that exogenous offshoring has a stronger negative effect on low-skilled wages than raw correlations imply.

8.4 Implications for inequality

The results imply that globalization increases wage inequality in several ways. It raises the skill premium within firms, creates variation in wage effects within the same education group, and imposes large losses on displaced workers. This means that distributional effects cannot be fully captured by comparing industries or education groups alone.

8.5 Policy implications

The paper suggests that adjustment policies should focus not only on industries exposed to imports but also on worker tasks and displacement risk. Low-skilled and routine workers face the largest downside risk. Export promotion and adjustment assistance may have very different distributional effects, because exporting raises labor demand while offshoring can reduce demand for specific tasks.

8.6 Limitations

The strongest causal evidence is for offshoring because the import instruments are closely tied to external supply and transport-cost shocks. Export instruments are useful but may be more exposed to correlated demand shocks. The paper also studies Denmark, a labor market with strong institutions and relatively high worker mobility support, so the magnitudes may differ in countries with weaker safety nets.

8.7 Takeaway

The paper’s lasting contribution is to show that the wage effects of offshoring are worker-specific, not just industry-specific. The relevant unit of analysis is the worker-firm-task match. Offshoring can raise firm

productivity and some workers' wages while lowering others' wages and earnings, especially for low-skilled and displaced workers.